

Day 8 - Linux to Linux Connection (SSH with Password)

Date: 06-10-2026 | Class: ISC2 | Notes with terminal screenshots

1. Topics covered today

- **Telnet** - a client used to establish a TCP connection to a remote host/port and interact with the server; traditionally used for remote terminal access (no security, plain text).
- **Netcat (nc)** - a general-purpose network utility that can create TCP/UDP connections, listen on ports, send/receive data and troubleshoot network connectivity. Example: `nc www.ssmit.ac.in`
- **Nmap** - a network discovery and security auditing tool used to discover hosts, scan ports, identify services and perform network reconnaissance.
- **nslookup** - gives the IP address of a domain name.
- **DNS / Domain / Subdomain:** a domain name is the human-readable address (e.g. `www.ssmit.ac.in`). "ac.in" = Indian academic/educational domain, "ssmit" = institution domain, "www" = web server. A child domain (subdomain) is a domain created as part of a main (parent) domain.
- **Data security - Linux to Linux data transfer** using SSH. Two modes: (1) **Password mode** - login with a username and password; (2) **Password-less mode** - login using cryptographic keys (tomorrow).

2. Telnet, Netcat and Nmap

Telnet

A client used to establish a TCP connection to a remote host/port and interact with the server; traditionally used for remote terminal access. It sends everything (including passwords) in plain text with no encryption, so SSH replaced it for logins. Today it is mainly used to test whether a port is open.

```
telnet <host> <port>
telnet www.ssmit.ac.in 80      # connect to a web server port
telnet 65.2.83.52 22          # shows the SSH banner if port 22 is open
# exit: press Ctrl + ] then type quit
sudo dnf install telnet      # install on Amazon Linux 2023
```

Netcat (nc)

A general-purpose network utility that can create TCP/UDP connections, listen on ports, send and receive data, and troubleshoot network connectivity.

```
nc www.ssmit.ac.in 80          # connect to a host/port
nc -zv 65.2.83.52 22          # check if a port is open (-z scan, -v verbose)
nc -l 9000                    # listen on port 9000 (receiver)
nc 65.2.83.52 9000            # connect to that listener (sender)
# send a file: receiver nc -l 9000 > file.txt
# sender nc <receiver-IP> 9000 < file.txt
sudo dnf install nmap-ncat    # install on Amazon Linux 2023
```

Nmap

A network discovery and security auditing tool used to discover hosts, scan ports, identify services and perform network reconnaissance. Only scan machines that you own or have written permission to test.

```
nmap 65.2.83.52          # scan the common ports of one host
nmap -p 22,80 65.2.83.52 # scan only specific ports
nmap -sV 65.2.83.52      # detect service names and versions
nmap -sn 172.31.0.0/24    # find live hosts in a network (no port scan)
sudo dnf install nmap     # install on Amazon Linux 2023
```

Tool	What it does	Typical use	Note
telnet	TCP client to a host/port	Test a port, old remote login	Plain text - not secure
nc (netcat)	Read/write network connections, listen on ports	Port check, send data/files, debugging	Plain text - not encrypted
nmap	Discover hosts, scan ports, identify services	Network scanning and security auditing	Scan only with permission
nslookup	Gives the IP address of a domain name	DNS lookup	Example: nslookup www.ssmi.ac.in

Try it with your two EC2 machines: `nmap -p 22 65.2.83.52` or `nc -zv 65.2.83.52 22` should show port 22 (SSH) as open, which means the connection can reach the SSH server, and any "Permission denied" comes from login settings, not the network.

3. Steps to connect Linux to Linux using a password

Do these steps on the machine you want to connect **TO** (the server / target machine).

```
# 1. (optional) set a hostname and refresh the shell
sudo hostnamectl set-hostname <name>
exec bash
```

```
# 2. create the user and set its password
sudo useradd somesha
sudo passwd somesha
```

```
# 3. (optional, only if the user needs sudo power)
sudo visudo          # or: sudo vim /etc/sudoers
somesha ALL=(ALL) NOPASSWD: ALL
```

```
# 4. allow password login in the SSH server
sudo vim /etc/ssh/sshd_config
:set nu          # show line numbers
line 65: PasswordAuthentication yes
:wq!            # save and exit
```

```
# 5. RESTART the SSH service (very important)
sudo systemctl restart sshd
```

```
# 6. find the public IP of this machine
sudo curl ifconfig.me
```

On the machine you connect **FROM** (the client machine):

```
ssh somesha@<IP of the target machine>
# type the password when asked
```

Useful helper commands from the notes: `w` (who is logged in), `w -i` (show IP instead of hostname), `w --help`, `cat /etc/passwd` (list of users), `hostname -I` (private IP).

4. Why "Permission denied" is coming - and is the code working?

Short answer: the commands you typed are mostly correct, but the setup is not finished, so the login does not work yet. Two things are wrong, and both are visible in your screenshots.

Problem 1 - The SSH service was never restarted

You typed `restart sshd.service` and the terminal replied `-bash: restart: command not found` (on both machines). "restart" is not a Linux command by itself. Because the restart failed, the SSH server is still running with the OLD settings, where password login is OFF (the Amazon Linux default is `PasswordAuthentication no`). Your own screenshot proves it: the error says `Permission denied (publickey,gssapi-keyex,gssapi-with-mic)` - only key-based methods are offered and **"password" is missing from the list**. That is why it never asks you to enter a password.

Correct command:

```
sudo systemctl restart sshd
# or the way written in your notes:
sudo service sshd restart
```

Problem 2 - Username does not match the machine

A user exists only on the machine where you created it. On the right-side machine (65.2.83.52) you created the user **kiran**, but from the left machine you tried `ssh somesha@65.2.83.52`. User "somesha" was not created on that machine. In the same way, from the right machine you tried `ssh kiran@52.66.243.206`, but "kiran" does not exist on the left machine (52.66.243.206).

Machine	Public IP	User created there?
Left (ip-172-31-13-95)	52.66.243.206	No new user created in this session
Right (ip-172-31-7-230)	65.2.83.52	kiran (useradd + passwd done)

Notes: (a) "BAD PASSWORD: shorter than 8 characters" is only a warning for root - the password was still saved, but use 8+ characters. (b) Editing `/etc/sudoers` is not required for SSH login; it only gives the user sudo power. (c) Editing `sshd_config` on the machine you connect FROM is not needed.

5. Fix - do these in order

```
# ON THE TARGET MACHINE (example: right machine, 65.2.83.52)
sudo vim /etc/ssh/sshd_config # PasswordAuthentication yes (no # in front)
sudo systemctl restart sshd   # must give NO error
sudo sshd -T | grep -i passwordauth # must print: passwordauthentication yes
```

```
# ON THE OTHER MACHINE (left machine)
ssh kiran@65.2.83.52 # user = the one created on the target
# now it should ask: kiran@65.2.83.52's password:
```

If it still shows the same error, check the folder `/etc/ssh/sshd_config.d/` on the target. A file there (for example `50-cloud-init.conf`) can override the setting with `PasswordAuthentication no`; change it to yes and restart sshd again. To connect in the opposite direction, create a user on the left machine first, then run ssh from the right machine.

6. Tomorrow's class

We learned today how to connect one Linux machine to another Linux machine using a **password**. In tomorrow's class we will learn how to connect a Linux machine **without a password**, by using an encrypted key pair (public/private key authentication).

7. Terminal screenshots (both Linux machines)

```
ec2-user@ip-172-31-13-95:~$ ssh -i "C:\Users\ksome\Downloads\SOMU86.pem" ec2-user@52.66.243.206
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Oct 6 14:56:22 2026 from 13.233.177.4
[ec2-user@ip-172-31-13-95 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-13-95 ~]$ sudo systemctl restart sshd.service
-bash: restart: command not found
[ec2-user@ip-172-31-13-95 ~]$ sudo curl ifconfig.me
52.66.243.206[ec2-user@ip-172-31-13-95 ~]$ w
15:10:27 up 19 min, 1 user, load average: 0.03, 0.02, 0.00
USER      TTY      LOGIN@   IDLE   JCPU   PCPU   WHAT
ec2-user pts/0    15:03    1.00s  0.00s  0.00s  w
[ec2-user@ip-172-31-13-95 ~]$ w -i
15:10:38 up 20 min, 1 user, load average: 0.03, 0.02, 0.00
USER      TTY      FROM      LOGIN@   IDLE   JCPU   PCPU   WHAT
ec2-user pts/0    157.50.191.2 15:03   4.00s  0.01s  0.00s  w -i
[ec2-user@ip-172-31-13-95 ~]$ ssh somesha@65.2.83.52
The authenticity of host '65.2.83.52 (65.2.83.52)' can't be established.
ED25519 key fingerprint is SHA256:D+wQDcZdEfGhD85L9X6ju6dZDEniUPAdF+aJtH/4tQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '65.2.83.52' (ED25519) to the list of known hosts
somesha@65.2.83.52: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-172-31-13-95 ~]$ w -i
15:12:48 up 22 min, 1 user, load average: 0.00, 0.01, 0.00
USER      TTY      FROM      LOGIN@   IDLE   JCPU   PCPU   WHAT

ec2-user@ip-172-31-7-230:~$ ssh -i "C:\Users\ksome\Downloads\SOMU86.pem" ec2-user@65.2.83.52
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-7-230 ~]$ sudo useradd kiran
[ec2-user@ip-172-31-7-230 ~]$ sudo passwd kiran
Changing password for user kiran.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-7-230 ~]$ sudo vim /etc/sudoers
[ec2-user@ip-172-31-7-230 ~]$ sudo vim /etc/ssh/sshd_config
[ec2-user@ip-172-31-7-230 ~]$ sudo systemctl restart sshd.service
-bash: restart: command not found
[ec2-user@ip-172-31-7-230 ~]$ ssh kiran@52.66.243.206
The authenticity of host '52.66.243.206 (52.66.243.206)' can't be established.
ED25519 key fingerprint is SHA256:S5exlc1fXYpK5kO/Wgb9WuBjEP3XU5gqQrW4T/krRo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.66.243.206' (ED25519) to the list of known hosts
kiran@52.66.243.206: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-172-31-7-230 ~]$ sudo curl ifconfig.me
65.2.83.52[ec2-user@ip-172-31-7-230 ~]$ w
15:12:03 up 21 min, 1 user, load average: 0.04, 0.03, 0.00
USER      TTY      LOGIN@   IDLE   JCPU   PCPU   WHAT
```

Left: 52.66.243.206 (sshd restart failed, ssh somesha@65.2.83.52 denied). Right: 65.2.83.52 (user kiran created, restart failed, ssh kiran@52.66.243.206 denied).

```
ec2-user@ip-172-31-13-95:~  
#/ --- https://aws.amazon.com/linux/amazon-linux-2023  
Vw! i-->  
m/!  
Last login: Tue Oct 6 14:56:22 2026 from 13.233.177.4  
[ec2-user@ip-172-31-13-95 ~]$ sudo vim /etc/ssh/sshd_config  
[ec2-user@ip-172-31-13-95 ~]$ restart sshd.service  
-bash: restart: command not found  
[ec2-user@ip-172-31-13-95 ~]$ sudo curl ifconfig.me  
52.66.243.206[ec2-user@ip-172-31-13-95 ~]$ w  
15:10:27 up 19 min, 1 user, load average: 0.03, 0.02, 0.00  
USER TTY LOGIN@ IDLE JCPU PCPU WHAT  
ec2-user pts/0 15:03 1.00s 0.01s 0.00s w  
[ec2-user@ip-172-31-13-95 ~]$ w -i  
15:10:38 up 20 min, 1 user, load average: 0.03, 0.02, 0.00  
USER TTY FROM LOGIN@ IDLE JCPU PCPU WHAT  
ec2-user pts/0 157.50.191.2 15:03 4.00s 0.01s 0.00s w -i  
[ec2-user@ip-172-31-13-95 ~]$ ssh somesha@65.2.83.52  
The authenticity of host '65.2.83.52 (65.2.83.52)' can't be established.  
ED25519 key fingerprint is SHA256:D+mqDcZdEfHghD85l9X6ju6dZDeniUPAdF+aJtH/4tQ.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '65.2.83.52' (ED25519) to the list of known hosts  
somesha@65.2.83.52: Permission denied (publickey,gssapi-keyex,gssapi-with-mi  
c).  
[ec2-user@ip-172-31-13-95 ~]$ w -i  
15:12:48 up 22 min, 1 user, load average: 0.00, 0.01, 0.00  
USER TTY FROM LOGIN@ IDLE JCPU PCPU WHAT  
ec2-user pts/0 157.50.191.2 15:03 0.00s 0.01s 0.00s w -i  
[ec2-user@ip-172-31-13-95 ~]$ w  
15:13:04 up 22 min, 1 user, load average: 0.00, 0.01, 0.00  
USER TTY LOGIN@ IDLE JCPU PCPU WHAT  
ec2-user pts/0 15:03 0.00s 0.01s 0.00s w  
[ec2-user@ip-172-31-13-95 ~]$ ssh somesha@65.2.83.52  
somesha@65.2.83.52: Permission denied (publickey,gssapi-keyex,gssapi-with-mi  
c).  
[ec2-user@ip-172-31-13-95 ~]$  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[ec2-user@ip-172-31-7-230 ~]$ sudo vim /etc/sudoers  
[ec2-user@ip-172-31-7-230 ~]$ sudo vim /etc/ssh/sshd_config  
[ec2-user@ip-172-31-7-230 ~]$ restart sshd.service  
-bash: restart: command not found  
[ec2-user@ip-172-31-7-230 ~]$ ssh kiran@52.66.243.206  
The authenticity of host '52.66.243.206 (52.66.243.206)' can't be establish  
ED25519 key fingerprint is SHA256:S5exlc+lfXYpK5k0/Wgb9WuBjEP3XU5gqQrM4T/knR  
o.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '52.66.243.206' (ED25519) to the list of known ho  
sts.  
kiran@52.66.243.206: Permission denied (publickey,gssapi-keyex,gssapi-with-m  
ic).  
[ec2-user@ip-172-31-7-230 ~]$ sudo curl ifconfig.me  
65.2.83.52[ec2-user@ip-172-31-7-230 ~]$ w  
15:12:03 up 21 min, 1 user, load average: 0.04, 0.03, 0.00  
USER TTY LOGIN@ IDLE JCPU PCPU WHAT  
ec2-user pts/0 14:57 1.00s 0.02s 0.00s w  
[ec2-user@ip-172-31-7-230 ~]$ w --help  
Usage:  
w [options]  
Options:  
-h, --no-header do not print header  
-u, --no-current ignore current process username  
-s, --short short format  
-f, --from show remote hostname field  
-o, --old-style old style output  
-i, --ip-addr display IP address instead of hostname (if possible)  
--help display this help and exit  
-V, --version output version information and exit  
For more details see w(1).  
[ec2-user@ip-172-31-7-230 ~]$ |
```

Later state: "Permission denied (publickey,gssapi-keyex,gssapi-with-mic)" - password is not offered because sshd was not restarted.

8. After the fix - password login enabled and sshd restarted

```
ec2-user@ip-172-31-13-95:~$ sudo systemctl restart sshd && sudo systemctl status sshd --no-pager
Last login: Tue Oct 6 15:03:51 2026 from 157.50.191.2
[ec2-user@ip-172-31-13-95 ~]$ sudo passwd somesha && sudo tee /etc/ssh/sshd_config.d/99-password-auth.conf > /dev/null <<'EOF'
PasswordAuthentication yes
KbdInteractiveAuthentication yes
UsePAM yes
EOF
sudo sshd -t && sudo systemctl restart sshd && sudo systemctl status sshd --no-pager
Changing password for user somesha.
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 15:22:50 UTC; 21ms ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 4438 (sshd)
    Tasks: 1 (limit: 9300)
   Memory: 1.3M
      CPU: 5ms
   CGroup: /system.slice/sshd.service
           └─4438 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup..."

Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal systemd[1]: Sta...
Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal sshd[4438]: Ser...
Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal sshd[4438]: Ser...
Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal systemd[1]: Sta...
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@ip-172-31-13-95 ~]$ w
 15:23:43 up 33 min,  2 users,  load average: 0.00, 0.00, 0.00
USER      TTY      LOGIN@  IDLE   JCPU   PCPU   WHAT
ec2-user pts/0    15:03  10:07   0.01s  0.01s  -bash
ec2-user pts/1    15:22  1.00s   0.01s  0.00s  w
[ec2-user@ip-172-31-13-95 ~]$
```

```
ec2-user@ip-172-31-7-230:~$ https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Oct 6 14:57:02 2026 from 157.50.191.2
[ec2-user@ip-172-31-7-230 ~]$ w
 15:23:13 up 32 min,  2 users,  load average: 0.00, 0.00, 0.00
USER      TTY      LOGIN@  IDLE   JCPU   PCPU   WHAT
ec2-user pts/0    14:57  10:57   0.02s  0.02s  -bash
ec2-user pts/1    15:23  1.00s   0.01s  0.00s  w
[ec2-user@ip-172-31-7-230 ~]$ sudo passwd somesha && sudo tee /etc/ssh/sshd_config.d/99-password-auth.conf > /dev/null <<'EOF'
PasswordAuthentication yes
KbdInteractiveAuthentication yes
UsePAM yes
EOF
sudo sshd -t && sudo systemctl restart sshd && sudo systemctl status sshd --no-pager
passwd: Unknown user name 'somesha'.
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 15:23:36 UTC; 21ms ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 4069 (sshd)
    Tasks: 1 (limit: 9300)
   Memory: 1.4M
      CPU: 5ms
   CGroup: /system.slice/sshd.service
           └─4069 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup..."

Oct 06 15:23:36 ip-172-31-7-230.ap-south-1.compute.internal systemd[1]: Sta...
Oct 06 15:23:36 ip-172-31-7-230.ap-south-1.compute.internal sshd[4069]: Ser...
Oct 06 15:23:36 ip-172-31-7-230.ap-south-1.compute.internal sshd[4069]: Ser...
Oct 06 15:23:36 ip-172-31-7-230.ap-south-1.compute.internal systemd[1]: Sta...
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@ip-172-31-7-230 ~]$
```

Both machines: PasswordAuthentication yes written to /etc/ssh/sshd_config.d/99-password-auth.conf, sshd config test passed, and sshd restarted successfully (active/running). Left machine: user "somesha" exists (password set). Right machine: "Unknown user name 'somesha'" - confirms that user is only on the left machine.

```
ec2-user@ip-172-31-13-95:~$ sudo sshd -t && sudo systemctl restart sshd && sudo systemctl status sshd --no-pager
Changing password for user somesha.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 15:22:50 UTC; 21ms ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 4438 (sshd)
    Tasks: 1 (limit: 9300)
   Memory: 1.3M
      CPU: 5ms
   CGroup: /system.slice/sshd.service
           └─4438 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup..."

Last login: Tue Oct 6 15:03:51 2026 from 157.50.191.2
[ec2-user@ip-172-31-13-95 ~]$ sudo passwd somesha && sudo tee /etc/ssh/sshd_config.d/99-password-auth.conf > /dev/null <<'EOF'
PasswordAuthentication yes
KbdInteractiveAuthentication yes
UsePAM yes
EOF
sudo sshd -t && sudo systemctl restart sshd && sudo systemctl status sshd --no-pager
Changing password for user somesha.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 15:22:50 UTC; 21ms ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 4438 (sshd)
    Tasks: 1 (limit: 9300)
   Memory: 1.3M
      CPU: 5ms
   CGroup: /system.slice/sshd.service
           └─4438 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup..."

Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal systemd[1]: Sta...
Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal sshd[4438]: Ser...
Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal sshd[4438]: S...
Oct 06 15:22:50 ip-172-31-13-95.ap-south-1.compute.internal systemd[1]: Sta...
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@ip-172-31-13-95 ~]$
```

Same session scrolled up (also shows the reconnect to the right machine and "w" with 2 users logged in).